第 7 週專題期末進度報告

學號：513558006

專題名稱：BirdEye-SQL（MSSQL 雙向 SQL <-> AST 引擎）

1. GitHub 活動紀錄

URL:https://github.com/wbird0606/BirdEye-SQL.git

1.1 貢獻摘要

- 專案總提交數（主分支歷史）：78
- 作者提交統計：
  - wbird0606：74 次提交
  - wbird：4 次提交

1.2 本週重點提交（第 7 週相關）

- 71bfa00（2026-04-03）：更新文件與 demo 匯出流程
- f3b5466（2026-04-03）：精簡影片展示段落
- e6b4511（2026-04-03）：將 demo 資產轉為網頁相容格式
- e515a77（2026-04-03）：效能優化（json -> ujson 與 schema cache）
- 3ff06ea（2026-03-24）：多 Schema 支援、新語法節點與覆蓋率補齊
- ba27656（2026-03-24）：新增 final coverage 測試，覆蓋率由 97% 提升至 99%
- 50b9abe（2026-03-23）：COUNT(*) / SELECT * 展開為逐欄 READ intent
- 54da04e（2026-03-23）：修正 IntentExtractor 邊界案例（#67-#73）
- 7efbc9c（2026-03-22）：新增 Web AST -> SQL 功能

- 總提交

Activity

All branches | All activity | wbird0606 | All time | Showing most recent first

- feat: implement validation for unclosed symbols, Closes #20
  wbird0606 pushed 1 commit to `main` • 66bc41e...c18cc8c • 16 days ago
- feat: enforce strict alias scope and support star expansion, Closes #19
  wbird0606 pushed 1 commit to `main` • b32b0ce...66bc41e • 16 days ago
- feat: support string/numeric literals and column aliases, Closes #18
  wbird0606 pushed 1 commit to `main` • 937f00a...b32b0ce • 16 days ago
- feat: implement SQL comment skipping for single and multi-line, Closes ...
  wbird0606 pushed 1 commit to `main` • f0f1c55...937f00a • 16 days ago
- feat: support identifiers with spaces in brackets, Closes #16
  wbird0606 pushed 1 commit to `main` • b46a7d9...f0f1c55 • 16 days ago
- test: ensure robust alias resolution across extreme whitespace and mi...
  wbird0606 pushed 1 commit to `main` • cbf12be...b46a7d9 • 16 days ago
- test: verify strict comma validation and trailing garbage defense, Cl...
  wbird0606 pushed 1 commit to `main` • 51739c9...cbf12be • 16 days ago
- feat: handle escaped keywords in brackets and fix symbol recognition, C...
  wbird0606 pushed 1 commit to `main` • cf360c9...51739c9 • 16 days ago
- feat: support multi-level identifiers and maintain backward compatibi...
  wbird0606 pushed 1 commit to `main` • 9ac10e8...cf360c9 • 16 days ago
- test: synchronize all test suites with new Parser/Binder logic, Closes ...
  wbird0606 pushed 1 commit to `main` • d41bd0b...9ac10e8 • 16 days ago
- chore: setup GitHub Actions for automated testing, Closes #8
  wbird0606 pushed 1 commit to `main` • f7cf713...d41bd0b • 16 days ago
- fix: resolve SQL injection vulnerability & expand lexer symbols, Closes
  wbird0606 pushed 1 commit to `main` • 2a4b165...f7cf713 • 16 days ago
- feat: implement semantic binder with star expansion, Closes #4
  wbird0606 pushed 1 commit to `main` • 2eb565a...2a4b165 • 16 days ago
- feat: implement recursive descent parser for SELECT statements, Closes ...
  wbird0606 pushed 1 commit to `main` • 56927bd...2eb565a • 16 days ago
- feat: implement zero-copy lexer, Closes #2
  wbird0606 pushed 1 commit to `main` • 46259f7...56927bd • 16 days ago
- Merge branch 'main' of https://github.com/wbird0606/BirdEye-SQL
  wbird0606 pushed 2 commits to `main` • dd30b70...46259f7 • 16 days ago
- Revise README for BirdEye-SQL project detailsupdate README.md
  wbird0606 pushed 1 commit to `main` • 7f19ff2...dd30b70 • 16 days ago
- Initial commit
  wbird0606 created `main` • 7f19ff2 • 16 days ago

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Activity

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- fix: 補齊 Visualizer 與 Serializer 對 DECLARE / SELECT INTO / APPLY 的支援
  wbird0606 pushed 1 commit to `main` • 250fe28...092c94b • 12 days ago
- feat: 擴充進階型別家族 (SPATIAL/BINARY/XML) 與 JSON/UUID 函數 (close #54)
  wbird0606 pushed 1 commit to `main` • d88ac9d...250fe28 • 12 days ago
- feat: 支援 CROSS/OUTER APPLY 關聯運算子與橫向作用域 (close #53)
  wbird0606 pushed 1 commit to `main` • 073cbd4...d88ac9d • 12 days ago
- feat: 支援 DECLARE 變數宣告、GO 批次分隔符與 #臨時表 (close #51, close #52)
  wbird0606 pushed 1 commit to `main` • e6d1e46...073cbd4 • 12 days ago
- feat: 補齊視覺化支援並達成 235 個 TDD 測試全數通過
  wbird0606 pushed 1 commit to `main` • 7845953...e6d1e46 • 12 days ago
- feat: 支援公用表表達式 (CTE / WITH) 與全系統語意架構優化
  wbird0606 pushed 1 commit to `main` • 9926e1f...7845953 • 13 days ago
- feat: 支援 SQL CAST/CONVERT 轉型與 UNION 集合運算
  wbird0606 pushed 1 commit to `main` • d23f4f9...9926e1f • 13 days ago
- feat: 支援 SQL BETWEEN 與 NOT BETWEEN 語法及型別驗證
  wbird0606 pushed 1 commit to `main` • b795e4f...d23f4f9 • 13 days ago
- chore: 移除過時的工具程式碼並清理專案結構

wbird0606 pushed 1 commit to main • e66dac6...b79e4f • 13 days ago

chore: 從版本控制中移除 __pycache__ 與 .pyc 檔案並更新 .gitignore
wbird0606 pushed 1 commit to main • 2c353dd...e66dac6 • 13 days ago

test: 更新尾隨逗號(Trailing Comma)邊界防禦的預期錯誤訊息
wbird0606 pushed 1 commit to main • 8765bc2...2c353dd • 13 days ago

docs: 精確更新 TDD 測試案例數量至 135+，並同步中英文版本
wbird0606 pushed 1 commit to main • 5a201c2...8765bc2 • 13 days ago

docs: 更新 README 支援中英雙語，並詳述 80+ TDD 測試案例與最新架構
wbird0606 pushed 1 commit to main • 428e3c6...5a201c2 • 13 days ago

feat: 全域引擎整合、CLI、Web UI 實作與語意防禦強化
wbird0606 pushed 1 commit to main • 2299f56...428e3c6 • 13 days ago

feat(semantic): implement strict ZTA validation and deep scope backtr...
wbird0606 pushed 1 commit to main • 006a170...2299f56 • 14 days ago

feat: add AST JSON Serializer and Mermaid exporter for visualization, c...
wbird0606 pushed 1 commit to main • 151d9be...006a170 • 14 days ago

feat: implement scalar function support, dynamic registry, and ZTA sa...
wbird0606 pushed 1 commit to main • 71f6f48...151d9be • 15 days ago

feat: implement CASE WHEN logic with support for simple/searched synt...
wbird0606 pushed 1 commit to main • f1a4e6e...71f6f48 • 15 days ago

refactor: implement recursive scope stack for nested subqueries and c...
wbird0606 pushed 1 commit to main • 98501c0...f1a4e6e • 15 days ago

feat: implement GROUP BY and HAVING with ZTA-compliant aggregate vali...
wbird0606 pushed 1 commit to main • f37e22d...98501c0 • 15 days ago

feat: implement TOP and ORDER BY clauses with ZTA-compliant semantic ...
wbird0606 pushed 1 commit to main • 20a28ac...f37e22d • 15 days ago

fixed bugs:add tests and fix more bugs
wbird0606 pushed 1 commit to main • b459617...20a28ac • 15 days ago

feat: complete DML visualization and final ZTA logic fixes, closes #28 ...
wbird0606 pushed 1 commit to main • 6bcc170...b459617 • 15 days ago

feat: implement UPDATE and DELETE with mandatory WHERE protection, cl...
wbird0606 pushed 1 commit to main • 1d07f84...6bcc170 • 15 days ago

fix: support float literals in lexer and complete Issue #29 visualizer
wbird0606 pushed 1 commit to main • 585115c...1d07f84 • 15 days ago

feat: implement nullability awareness for outer joins, closes #26
wbird0606 pushed 1 commit to main • d46aa23...585115c • 15 days ago

feat: implement incremental scope registration for Issue #25
wbird0606 pushed 1 commit to main • 99e703f...d46aa23 • 15 days ago

feat/sec: implement expression engine and function sandbox with ZTA w...
wbird0606 pushed 1 commit to main • 7958b4b...99e703f • 15 days ago

feat/sec: implement multi-table JOIN with ZTA ambiguity checks and ce...
wbird0606 pushed 1 commit to main • c0780d0...7958b4b • 15 days ago

security: implement nested comments and bracket escaping (fixes #21, f...
wbird0606 pushed 1 commit to main • c18cc8c...c0780d0 • 15 days ago

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Activity

All branches | All activity | wbird0606 | All time | Showing most recent first

docs: refresh demo assets and simplify export flow
wbird0606 pushed 1 commit to main • f3b5466...71bfa00 • 3 hours ago

refactor: simplify video demo section with MP4 download link only
wbird0606 pushed 1 commit to main • e6b4511...f3b5466 • 4 hours ago

refactor: convert demo assets to web-compatible formats (MP4 and PNG)
wbird0606 pushed 1 commit to main • 76745f6...e6b4511 • 4 hours ago

Merge branch 'feat/zta-proxy-integration'
wbird0606 pushed 2 commits to main • d0d8c89...76745f6 • 4 hours ago

Merge pull request #77 from wbird0606/feat/zta-proxy-integration Pull request merge
wbird0606 pushed 2 commits to main • 1a7a5ac...d0d8c89 • 4 hours ago

優化效能(json->ujson以及建立schema cache)
wbird0606 pushed 1 commit to feat/zta-proxy-integration • 5250d9e...e515a77 • 4 hours ago

Merge pull request #76 from wbird0606/feat/zta-proxy-integration Pull request merge
wbird0606 pushed 8 commits to main • 7efbc9c...1a7a5ac • 8 days ago

docs: 更新 README — 864 tests、99% 覆蓋率、新增 schema 匯出教學與 /api/intent 端點
wbird0606 pushed 1 commit to feat/zta-proxy-integration • ba27656...5250d9e • 8 days ago

test: 新增 test_final_coverage_suite — 覆蓋率從 97% 提升至 99%
wbird0606 pushed 1 commit to feat/zta-proxy-integration • 3ff06ea...ba27656 • 9 days ago

feat: 多 Schema 支援 + 新語法節點 (IF/EXEC/MERGE/DDL) + 覆蓋率補齊 close #75
wbird0606 pushed 1 commit to feat/zta-proxy-integration • 7decd34...3ff06ea • 9 days ago

chore: 將 CLAUDE.md 加入 .gitignore
wbird0606 pushed 1 commit to feat/zta-proxy-integration • 50b9abe...7decd34 • 11 days ago

feat: COUNT(*)/SELECT * 展開為逐欄 READ intent close #74
wbird0606 pushed 1 commit to feat/zta-proxy-integration • 693b36a...50b9abe • 11 days ago

feat: visualizer 顯示 binder 解析的 resolved_table（非限定欄位來源）
wbird0606 pushed 1 commit to feat/zta-proxy-integration • 54da04e...693b36a • 11 days ago

feat: 修復 IntentExtractor edge cases #67-#73 Force push
wbird0606 force pushed to feat/zta-proxy-integration • 97ca0e3...54da04e • 11 days ago

feat: 修復 IntentExtractor edge cases #71-#73 — Derived table、DELETE 相關...
wbird0606 pushed 1 commit to feat/zta-proxy-integration • 44b62e2...97ca0e3 • 11 days ago

feat: 修復 IntentExtractor edge cases #67-#70 — 非限定欄位、多表 SELECT *、相關子查詢...
wbird0606 created feat/zta-proxy-integration • 44b62e2 • 11 days ago

新增web ast->SQL功能
wbird0606 pushed 1 commit to main • c67b7a8...7efbc9c • 12 days ago

docs: 更新 README 以反映雙向 SQL↔AST 功能與 533 個測試案例 close #66
wbird0606 pushed 1 commit to main • 4db3723...c67b7a8 • 12 days ago

feat: Issue #5 round 4-6 邊界修復、測試整合與 README 更新功能修復： - % 模數、& | ^ ~ 位元運...
wbird0606 pushed 1 commit to main • aaf13aa...4db3723 • 12 days ago

refactor: 整合測試套件 + 更新 README 測試重整 (43 → 19 個檔案，489 tests 零回歸)： - test...
wbird0606 pushed 1 commit to main • b4faf26...aaf13aa • 12 days ago

feat: 修復運算子支援並補齊多項 MSSQL 邊界案例 (Issue #5 round 4-5)- 新增 % 模數、& | ^ ~ 位...
wbird0606 pushed 1 commit to main • fb74593...b4faf26 • 12 days ago

feat: 支援 WITH CTE + UPDATE/DELETE (Refs #5) - AST: UpdateStatement / ...
wbird0606 pushed 1 commit to main • 6db4c48...fb74593 • 12 days ago

feat: 補齊內建函數registry與日期部分識別符處理 (Refs #5)- Registry: 新增 30+ 常用函數 - NUL...
wbird0606 pushed 1 commit to main • 69915a7...6db4c48 • 12 days ago

feat: 支援衍生資料表、MAX/MIN、N''前綴、JOIN子查詢 (Refs #5) - Registry: 新增 MAX / MI...
wbird0606 pushed 1 commit to main • 5e51c36...69915a7 • 12 days ago

feat: 支援 NOT IN / NOT EXISTS / CROSS JOIN / FULL OUTER JOIN / OFFSET ...
wbird0606 pushed 1 commit to main • dedcdd9...5e51c36 • 12 days ago

feat: 支援 TOP N PERCENT 語法 (#59) - Lexer: 新增 KEYWORD_PERCENT token - A...
wbird0606 pushed 1 commit to main • eb7d9fb...dedcdd9 • 12 days ago

feat: 支援 INSERT INTO...SELECT 與 Multi-row VALUES (#57 #58) - AST: Ins...
wbird0606 pushed 1 commit to main • 85e1df6...eb7d9fb • 12 days ago

feat: 支援 SELECT DISTINCT 與 NULL 字面值 (#55 #56) - Lexer: 新增 KEYWORD_DIS...
wbird0606 pushed 1 commit to main • b50b3e6...85e1df6 • 12 days ago

test: 新增 MSSQL 邊界測試並修正負數字面值解析 (Bug-1)- 新增 tests/test_mssql_boundary_s...
wbird0606 pushed 1 commit to main • 655612d...b50b3e6 • 12 days ago

test: 補齊跨 issue 整合測試與 Visualizer/Serializer 新節點支援 (307 tests)
wbird0606 pushed 1 commit to main • 092c94b...655612d • 12 days ago

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1.3 期間活動快照（程式碼變更面向）

- 核心模組大幅更新：
  - Parser、Binder、Registry、Reconstructor、Serializer、Visualizer
- 測試規模與穩定性提升：
  - 新增或更新 intent、parser coverage、reconstructor coverage、final coverage 測試
- 文件與 Web 介面持續優化：
  - README 更新與前端/API 內容調整

2. 目前已完成功能

2.1 核心引擎與流程

- 完成 SQL 全流程處理：
  - Lexer -> Parser -> Binder -> Visualizer/Serializer -> Reconstructor
- 完成雙向轉換：
  - SQL -> AST JSON
  - AST JSON -> SQL

2.2 MSSQL 語法支援

- 已實作並穩定化的語法包含：
  - TOP / TOP PERCENT
  - OFFSET / FETCH
  - WITH CTE（含 CTE + UPDATE/DELETE）
  - DECLARE、GO 批次處理、暫存資料表（# / ##）
  - CROSS APPLY / OUTER APPLY
  - 集合運算：UNION、INTERSECT、EXCEPT
  - 衍生資料表與巢狀子查詢

2.3 語意分析與安全（ZTA）

- 已實作嚴格零信任語意驗證：
  - 型別推論與相容性檢查
  - Alias 政策強制
  - 多表情境欄位歧義防禦
  - 函數白名單（sandbox）限制
  - 函數限制名單與危險函數封鎖
  - 危險預存程序黑名單（例如 XP_CMDSHELL、SP_EXECUTESQL 等）
- 新增欄位層級 intent 擷取：與 ZTA / 權限映射模組整合，需搭配相關上下文才能完整使用
  - READ / WRITE / DELETE 權限意圖映射

2.4 工具與介面

- CLI 支援雙向流程（SQL -> AST 與 AST -> SQL）
- Flask Web API 端點：
  - /api/parse
  - /api/reconstruct
  - /api/upload_csv
  - /api/intent：與 ZTA / 權限映射模組整合，需搭配相關上下文才能完整使用
- Web Dashboard 可進行互動式解析與重建

3. 已解決問題

3.1 語法解析與邊界修正

- 修正一元負號與負數字面值邊界案例
- 修正 CAST/CONVERT 在長度與 style 的解析行為
- 修正 keyword-as-alias 邊界情況
- 改善函數名稱與關鍵字重疊時的解析處理
- 修正 DISTINCT 與 NULL 字面值的語法解析

- 修正 TOP PERCENT 與 OFFSET/FETCH 分頁語法
- 修正 INSERT INTO...SELECT 與多列 VALUES 解析
- 修正 NOT IN / NOT EXISTS / CROSS JOIN / FULL OUTER JOIN 語法
- 修正 DECLARE、SELECT INTO、APPLY、MERGE、IF/ELSE 等新語法節點
- 修正 UNION / INTERSECT / EXCEPT 作為衍生資料表時的解析行為
- 修正巢狀衍生資料表與 JOIN 子查詢的 alias 要求
- 修正 BULK INSERT 與 GO 批次語法在腳本層的處理
- 修正 TRY_CAST / TRY_CONVERT 與多種字串函數的語法支援

3.2 語意與推論修正

- 修正純量/關聯子查詢 inferred_type 傳遞問題
- 改善外部連接（outer join）nullable 行為
- 修正集合運算衍生資料表的欄位名稱保留問題
- 修正 COUNT(DISTINCT) 與比較/集合運算的型別檢查
- 修正 JOIN、子查詢與衍生資料表的 scope / nullable 傳遞
- 修正暫存資料表（# / ##）與 GO 批次處理的狀態保留
- 修正日期部分識別符（YEAR / MONTH / DAY 等）與函數名稱衝突
- 修正 COUNT(*) / SELECT * 展開為逐欄 READ intent 的判定邏輯
- 修正 APPLY / 視窗函數 / set-op 邊界在 binder 中的解析與轉譯
- 修正函數白名單與內建函數註冊表的回傳型別/參數型別對應
- 修正 resolved_table 顯示與欄位來源追蹤的可視化資訊

3.3 效能與可維護性

- 導入 schema cache
- 優化 JSON 處理路徑（ujson）
- 重新整併測試套件結構並維持零回歸
- 整併並擴充覆蓋率導向測試套件
- 同步更新 Visualizer / Serializer 對 DECLARE、SELECT INTO、APPLY 等節點支援
- 新增與整併 intent、parser coverage、reconstructor coverage、final coverage 測試
- 壓縮並重整測試套件數量，維持功能覆蓋與執行穩定性

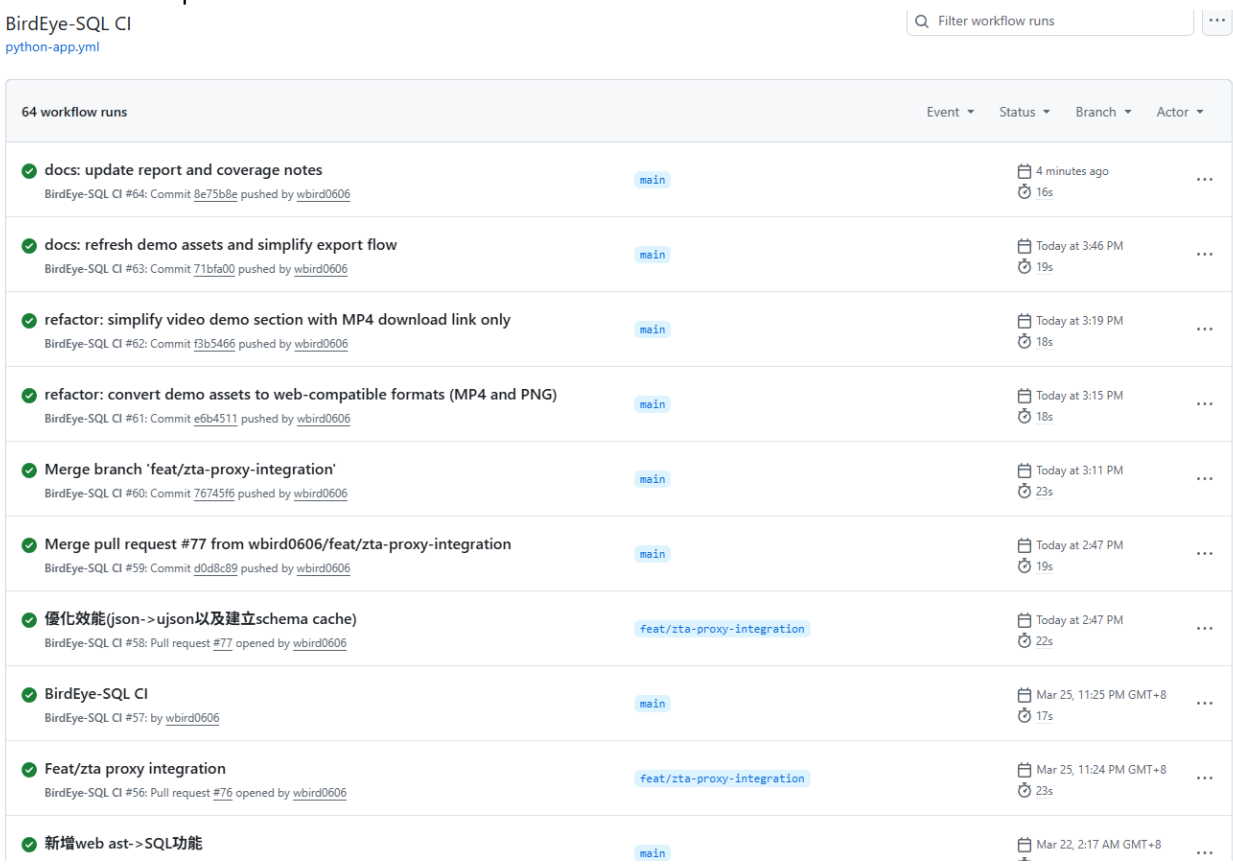
3.4 介面與整合修正

- 完成 Web AST -> SQL 重建流程
- 強化多 Schema 元數據支援與載入流程
- 修正 IntentExtractor 邊界案例與欄位層級 intent 擷取

4. 驗證與品質現況

- 目前測試結果：864/864 全數通過
- 目前行覆蓋率：100%
- TDD 原則：既有測試規格不變動，僅透過產品程式修正達成通過與覆蓋率目標
- 迴歸策略：每項功能搭配對應測試與整合測試，降低回歸風險

- 已建立CI，每次push皆會經過測試驗證



5. 後續規劃（期末展示前）

- 持續補強深層巢狀 MSSQL 語句壓力測試
- 強化 API 錯誤訊息與邊界防護
- 優化最終展示流程與範例敘事

6. 結論

截至第 7 週，專案已達成高穩定、高覆蓋率里程碑，具備完整 SQL <-> AST 雙向能力、嚴格語意/ZTA 防護，以及可用的 CLI/Web 操作介面。本週重點成果為效能優化、覆蓋率強化以及展示文件整理。